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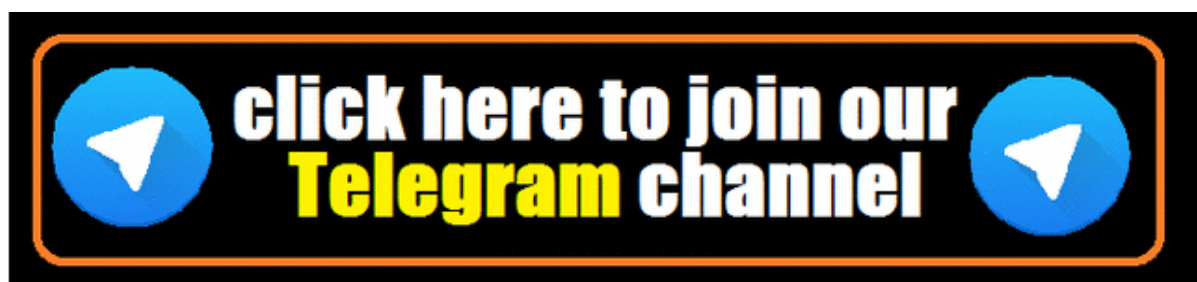
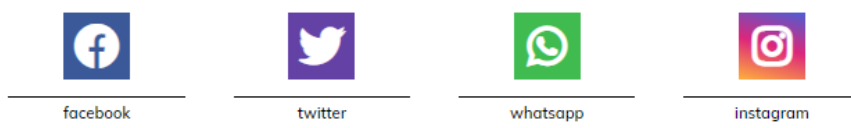


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IMPORTANT FACTS AND FORMULAE FOR JEE

AIEEE - MATHEMATICS

UNITS – 1: SETS, RELATIONS & FUNCTIONS

Algebraic properties of sets:

1. $A \cup (B \cap C) = (A \cup B) \cap C$
2. $A \cap B = B \cap A$
3. $A \cap \phi = \phi$
4. $A \subseteq B \Leftrightarrow A \cap B = A \Leftrightarrow A \cup B = B$
5. $(A \cap B)' = A' \cup B'$ and $(A \cup B)' = A' \cap B'$ **(De Morgan's Laws)**

Formulae for domains of functions:

1. $\text{dom}(f + g) = \text{dom} f \cap \text{dom} g$
2. $\text{dom}(fg) = \text{dom} f \cap \text{dom} g$
3. $\text{dom}(f/g) = \text{dom} f \cap \text{dom} g \cap \{x : g(x) \neq 0\}$
4. $\text{dom}(\sqrt{f}) = \text{dom} f \cap \{x : f(x) \geq 0\}$

Type of functions:

1. **Surjective function:** If a function $f : A \rightarrow B$ is such that each element in B is an image of atleast one element in A , then f is a function of A 'onto' B or f is a *surjective function* (also called *onto function*) from A to B .
2. **Injective function:** If function does not take the same value at two distinct points in its domain, then the function is said to be an *injective function* (also called *one-to-one function*)
3. **Bijjective function:** If a function f is both one-to-one and onto, the f is said to be a *bijjective function*.

UNITS - 2: COMPLEX NUMBERS

1. Property of Order

It states that $(a + ib) < (\text{or}) > c + id$ is not defined for complex numbers. For example, the statement $9 + 6i < 2 - i$ make no sense.

- i) The sum of four consecutive powers of i is zero.

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0, n \in \mathbb{I}$$

- ii) $\sqrt{a}\sqrt{b} = \sqrt{ab}$ is valid only if atleast one of a and b is non-negative.

If a and b are both negative then

$$\sqrt{a}\sqrt{b} = -\sqrt{ab}$$

2. Properties of Conjugate Complex Numbers

(i) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ and $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$

(ii) $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$ and $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$

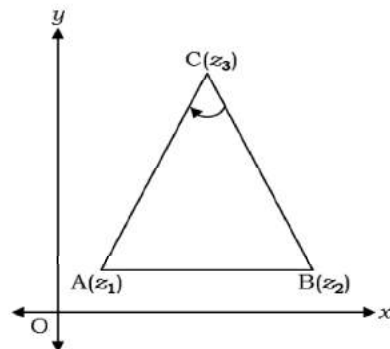
(iii) $z_1 \overline{z_2} - \overline{z_1} z_2 = 2 \operatorname{Re}(z_1 \overline{z_2})$

3. Rotational approach

If z_1, z_2, z_3 be vertices of a triangle ABC described in counter-clockwise sense (see Fig.) then:

$$\frac{(z_2 - z_3)}{|z_2 - z_3|} = \frac{(z_1 - z_3)}{|z_1 - z_3|} e^{i\alpha}$$

or $\operatorname{amp} \frac{z_2 - z_3}{z_1 - z_3} = \alpha = \angle BCA$



4. Properties of Modulus

(i) $|z| = |\overline{z}| = |-z| = |-\overline{z}|$

- (ii) $z\bar{z} = |z|^2$
- (iii) $|z^n| = |z|^n$
- (iv) $|z_1 + z_2| \leq |z_1| + |z_2|$
- (v) $|z_1 - z_2| \geq |z_1| - |z_2|$
- (vi) $|z_1 z_2| = |z_1| |z_2|$
- (vii) $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$
- (viii) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\{|z_1|^2 + |z_2|^2\}$
- (ix) $||z_1| - |z_2|| \leq |z_1 - z_2|$
- (x) $|az_1 - bz_2|^2 + |bz_1 + az_2|^2 = (a^2 + b^2)(|z_1|^2 + |z_2|^2)$, where $a, b \in \mathbb{R}$.

5. Properties of Argument of Complex Numbers

- (i) $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$
- (ii) $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2)$
- (iii) $\text{Arg}(z_1 z_2 z_3 \dots z_n) = \text{Arg}(z_1) + \text{Arg}(z_2) + \text{Arg}(z_3) \dots + \text{Arg}(z_n)$
- (iv) $\text{Arg}(z^n) = n \text{Arg}(z)$
- (v) If $\text{Arg}\left(\frac{z_2}{z_1}\right) = \theta$, then $\text{Arg} \frac{z_1}{z_2} = 2k\pi - \theta, k \in \mathbb{I}$

6. Demoivre's theorem

- (a) If n is a positive or negative integer, then

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

- (b) If n is a positive integer, then

$$(\cos \theta + i \sin \theta)^{1/n} = \cos\left(\frac{2k\pi + \theta}{n}\right) + i \sin\left(\frac{2k\pi + \theta}{n}\right)$$

Demoivre's theorem is valid if n is any rational number.

7. Some important Results

(i) If z_1 and z_2 are two complex numbers, then the distance between z_1 and z_2 is $|z_1 - z_2|$.

(ii) Segment joining points $A(z_1)$ and $B(z_2)$ is divided by point $P(z)$ in the ratio $m_1 : m_2$ then $z = \frac{m_1 z_2 + m_2 z_1}{(m_1 + m_2)}$, m_1 and m_2 are real.

(iii) The equation of the line joining z_1 and z_2 is given by

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0 \quad (\text{non-parametric form})$$

(iv) Three points z_1 , z_2 and z_3 are collinear if

$$\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = 0$$

(v) $\bar{a}z + a\bar{z} + b = 0$, $b \in \mathbb{R}$ describes equation of a straight line

Slope of line: The complex slope of the line $\bar{a}z + a\bar{z} + b = 0$ is

$$-\frac{a}{b} = \frac{\text{coeffi. of } \bar{z}}{\text{coeffi. of } z}$$

(vi) $|z - z_0| = r$ is equation of a circle, whose centre is z_0 and radius is r

(vii) If $|z - z_1| + |z - z_2| = 2a$, where $2a > |z_1 - z_2|$ then point z describes an ellipse.

(viii) $\frac{|z - z_1|}{|z - z_2|} = k$ is a circle if $k \neq 1$, and is a line if $k = 1$.

(ix) If $\text{Arg} \frac{(z_1 - z_3)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_3)} = \pm\pi, 0$, the points z_1, z_2, z_3, z_4 are concyclic.

UNIT – 3: MATRICES & DETERMINANTS

1. Some Important Terms

A matrix $A = (a_{ij})_{m \times n}$ is said to be a

- (i) Row matrix if $m = 1$
- (ii) Column matrix if $n = 1$
- (iii) Null or zero matrix if $a_{ij} = 0, \forall i \text{ and } j$
- (iv) Square matrix if $m = n$
- (v) Diagonal matrix if $m = n$ and $a_{ij} = 0, \forall i \neq j$
- (vi) Scalar matrix if $m = n$ and $a_{ij} = 0, \forall i \neq j$
and $a_{ij} = \lambda, \forall i = j$
- (vii) Unit or identity matrix if $m = n$ and $a_{ij} = 0, \forall i \neq j$
and $a_{ij} = 1, \forall i = j$

2. Properties of matrix multiplication

- i) If A is a square matrix, then

$$A^m A^n = A^{m+n}, \forall m, n \in \mathbb{N}$$

$$(A^m)^n = A^{mn}, \forall m, n \in \mathbb{N}$$
- ii) If A is an invertible matrix then

$$(A^{-1}BA)^m = A^{-1} B^m A$$
and $A^{-m} = (A^{-1})^m, \forall m \in \mathbb{N}$

3. Transpose of a Matrix

Definition: The transpose of a matrix $A = (a_{ij})_{m \times n}$, denoted by A'

(or by A^T) is the matrix $A' = (b_{ij})_{n \times m}$, where $b_{ij} = a_{ji}, \forall i \text{ and } j$.

By $\bar{A}$ we mean a matrix $B = (b_{ij})_{m \times n}$, where $b_{ij} = \bar{a}_{ij}$, where $\bar{a}$ denotes conjugate of a and by A^* we mean

$$A^* = (\bar{A})' = (\bar{A}^T)$$

Properties of Transpose of a Matrix

- i. $(A + B)' = A' + B'$
- ii. $(kA)' = kA'$ where k is a scalar.
- iii. $(AB)' = B'A'$ [Reversal law]
- iv. If A is an invertible matrix, then $(A^{-1})' = (A')^{-1}$

4. Adjoint and Inverse of a Matrix

The adjoint of a square matrix $A = (a_{ij})_{m \times n}$ is defined to be the matrix $\text{adj. } A = (b_{ij})_{n \times m}$, where $b_{ij} = A_{ji}$ where, A_{ji} is the cofactor of (j, i)th element of A.

Properties of Adjoint

- i) $A(\text{adj } A) = (\text{adj } A) A = |A| I_n$
- ii) $\text{adj } (kA) = k^{n-1} (\text{adj } A)$
- iii) $\text{adj}(AB) = (\text{adj } B) (\text{adj } A)$

Properties of Inverse

- i) $AA^{-1} = A^{-1}A = I_n$
- ii) $(A^{-1})^{-1} = A$
- iii) $(kA)^{-1} = k^{-1} A^{-1}$ if $k \neq 0$
- iv) $(AB)^{-1} = B^{-1}A^{-1}$ [reversal law]
- v) Inverse of a matrix if it exists is unique.
- vi) For a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\text{Adj. } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
and $A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ if $ad - bc \neq 0$

5. Properties of Determinants

- i) The interchange of any two rows (or columns) in Δ changes its sign.
- ii) If all the elements of a row (or column) in Δ are zero or if two rows (or columns) are identical (or proportional), then the value of Δ is zero.
- iii) If all the elements of one row (or column) is multiplied by a non-zero number k , then the value of the new determinant is k times the value of the original determinant.
- iv) If Δ becomes zero on putting $x = \alpha$, then we say that $(x - \alpha)$ is factor of Δ .

$$\text{v) } \begin{vmatrix} a_1 & a_2 & a_3 \\ 0 & b_2 & b_3 \\ 0 & 0 & c_3 \end{vmatrix} = a_1 b_2 c_3 = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & 0 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

6. Systems of Linear Equations

The system of homogeneous linear equations

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

and $a_3x + b_3y + c_3z = 0$

has a non-trivial solution (i.e., at least one of x, y, z is non-zero) if

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

and if $\Delta \neq 0$, then $x = y = z = 0$ is the only solution of above system (Trivial solution).

Cramer's Rule: Let us consider a system of equations

$$a_1x + b_1y + c_1z = d_1;$$

$$a_2x + b_2y + c_2z = d_2;$$

$$a_3x + b_3y + c_3z = d_3;$$

$$\text{Here } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix},$$

$$\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix},$$

By Cramer's Rule, we have $x = \frac{\Delta_1}{\Delta}$, $y = \frac{\Delta_2}{\Delta}$, and $z = \frac{\Delta_3}{\Delta}$

7. Differentiation of Determinant Function

If $F(x) = \begin{vmatrix} f_1 & f_1 & f_1 \\ g_2 & g_2 & g_2 \\ h_3 & h_3 & h_3 \end{vmatrix}$ where $f_1, f_2, f_3; g_1, g_2, g_3; h_1, h_2, h_3;$ are the functions of x , then

$$\therefore F'(x) = \begin{vmatrix} f'_1 & f'_1 & f'_1 \\ g_2 & g_2 & g_2 \\ h_3 & h_3 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_1 & f_1 \\ g'_2 & g'_2 & g'_2 \\ h_3 & h_3 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_1 & f_1 \\ g_2 & g_2 & g_2 \\ h'_3 & h'_3 & h'_3 \end{vmatrix}$$

8. Cyclic Order

If elements of the rows (or columns) are in cyclic order,

$$\text{i.e., i) } \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\text{ii) } \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$\text{iii) } \begin{vmatrix} a & bc & abc \\ b & ca & abc \\ c & ab & abc \end{vmatrix} = \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} = abc(a-b)(b-c)(c-a)$$

$$\text{iv) } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$\text{v) } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = (a^2 + b^2 + c^2 - 3abc)$$

UNIT - 4: QUADRATIC EQUATIONS

1. Roots of the equation

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ are } \alpha \text{ and } \beta$$

2. Sum and product of the roots:

$$S = \alpha + \beta = -\frac{b}{a}; \quad P = \alpha\beta = \frac{c}{a}.$$

3. To find the equation whose roots are α and β

$$x^2 - Sx + P = 0$$

where S is sum and P is product of roots.

4. Nature of the roots

$D = b^2 - 4ac$ where D is called discriminant.

(a) If $b^2 - 4ac \geq 0$, roots are real

i) $b^2 - 4ac > 0$, roots are real and unequal.

ii) $b^2 - 4ac = 0$, roots are real and equal. In this case, each

$$\text{root} = -\frac{b}{2a}$$

iii) $b^2 - 4ac = \text{perfect square}$, roots are rational.

iv) $b^2 - 4ac = \text{not a perfect square}$, roots are irrational.

(b) if $b^2 - 4ac < 0$, i.e., -ve, then $\sqrt{b^2 - 4ac}$ is imaginary.

Therefore the roots are imaginary and unequal.

5. Symmetric function of the roots

If α and β are the roots of $ax^2 + bx + c = 0$, then

- i) $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$
- ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- iii) $\alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$
- iv) $\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$
- v) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
- vi) $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$

6. Condition for Common roots

Equations: $a_1x^2 + b_1x + c_1 = 0$

Roots: $a_2x^2 + b_2x + c_2 = 0$

Condition for one common root:

$$(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$$

Condition for both the roots to be common:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

⇒ **Note:** If the coefficient of x^2 in both the equations be unity, then the value of common root is obtained by subtracting their equations.

7. Inequalities

- a) $x^2 > a^2$ is positive if $x < -a$ or $x > a$
- b) $x^2 < a^2$ if $-a < x < a$
- c) $a^2 < x^2 < b^2$ double inequality $\therefore a < x < b$ or $-b < x < -a$.

8. In an inequality you can always multiply or divide by a +ve quantity but not by a -ve quantity. Multiplying by a -ve quantity or taking reciprocal will reverse the inequality.

e.g., $a > b \Rightarrow -a < -b$ or $\frac{1}{a} < \frac{1}{b}$.

UNIT – 5: PERMUTATIONS & COMBINATIONS

1. Fundamental Principle of Counting

- (i) **Multiplication Principle:** There are m ways doing one work and n way doing another work then way of doing both work together
 $= m.n.$
- (ii) **Addition Principle:** There are m way doing one work and n ways doing another work then ways doing either m ways or n ways
 $= m + n.$

2. The number of ways of arranging n distinct objects in a row taking r ($0 \leq r \leq n$) at a time is denoted by ${}^n P_r$ or $P(n, r)$

$$\text{and } {}^n P_r = n(n-1)(n-2)\dots(n-r+1) = \frac{n!}{(n-r)!}$$

Note that, ${}^n P_0 = 1$, ${}^n P_1 = n$ and ${}^n P_{n-1} = {}^n P_n = n!$

3. The number of ways of arranging n distinct objects along a circle is $(n-1)!$
4. The number of ways of arranging n beads along a circular wire is $\frac{(n-1)!}{2}$.
5. The number of permutations of n things taken all at a time, p are alike of one kind, q are alike of another kind and r are alike of a third kind and the rest $n - (p + q + r)$ are all different is $\frac{n!}{p!q!r!}$.
6. The number of ways of arranging n distinct objects taking r of them at a time where any object may be repeated any number of times is n^r .
7. The number of selecting at least one object out of ' n ' distinct object
 $= 2^n - 1$
8. **Derangements:** The number of derangements (No object goes to its scheduled place) of n objects.

$$= n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!} \right)$$

9. The coefficient of x^r in the expansion of $(1-x)^{-n} = {}^{n+r-1}C_r$

10. Some Important Results

- (i) ${}^nC_0 = {}^nC_n = 1, {}^nC_1 = n$
- (ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
- (iii) ${}^nC_x = {}^nC_y \Leftrightarrow x = y \text{ or } x + y = n$
- (iv) ${}^{2n+1}C_0 + {}^{2n+1}C_1 + \dots + {}^{2n+1}C_n = 2^{2n}$
- (v) ${}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1}$
- (vi) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$
- (vii) ${}^nC_r + {}^{n+1}C_n + {}^{n+2}C_n + \dots + {}^{2n+1}C_n = {}^{2n}C_{n+1}$

UNIT – 6: MATHEMATICAL INDUCTION AND ITS APPLICATIONS

It is often used to prove a statement depending upon a natural number n .

Ist Principle of Induction

If $P(n)$ is a statement depending upon n , then to prove it by induction, we proceed as follow:

- i) Verify the validity of $P(n)$ for $n = 1$
- ii) Assume that $P(n)$ is true for some positive integer m and then using it establish the validity of $p(n)$ for $n = m + 1$.

Then, $P(n)$ is true for each $n \in \mathbb{N}$

IInd Principle of Induction:

In $P(n)$ is a statement depending upon n but beginning with some positive integer k , then to prove $P(n)$, we proceed as follows:

- i) Verify the validity of $P(n)$ for $n = k$
- ii) Assume that the statement is true for $n = m \geq k$. Then, using it establish the validity of $P(n)$ for $n = m + 1$.

Then, $P(n)$ is true for each $n \geq k$.

UNIT – 7: BINOMIAL THEOREM & ITS APPLICATIONS

1. Binomial Theorem

It n is a positive integer and $x, y, \in \mathbb{C}$ then

$$(x + y)^n = {}^nC_0 x^{n-0} y^0 + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_n y^n$$

Here ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are called binomial coefficient.

2. Some Important Points to Remember

- (i) **General term:** General term = $(r + 1)$ th term
 $\therefore \Rightarrow T_{r+1} = {}^nC_r x^{n-r} y^r$, where $r = 0, 1, 2, \dots, n$.
- (ii) **Middle term:** The middle term depends upon the value of n .
 - (a) If n is even,
 i.e., $\left(\frac{n}{2} + 1\right)$ th term is the middle term.
 - (b) If n is odd, there are two middle terms,
 i.e., $\left(\frac{n+1}{2}\right)$ th and $\left(\frac{n+3}{2}\right)$ th are two middle terms.
- (iii) **Greatest Term:**
 - (a) If p is an integer, then T_p and T_{p+1} are equal and both are greatest term
 - (b) If p is not an integer, then $T_{[p]+1}$ is the greatest term, where $[.]$ denotes the greatest integral part.
- (iv) **Greatest Coefficient:**
 - (a) If n is even, then greatest coefficient ${}^nC_{n/2}$
 - (b) If n is odd, then greatest coefficients are ${}^nC_{\frac{n-1}{2}}$ and ${}^nC_{\frac{n+1}{2}}$.
- (v) **Important Formulae:**
 - (a) $C_0 + C_1 + C_2 + C_3 + \dots + C_n = 2^n$
 - (b) $C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}$ $\Rightarrow$ sum of odd binomial coefficients = sum of even binomial coefficients
 - (c) $C_0 \quad C_1^2 \quad C_2^2 \quad \dots \quad C_n^2 \quad 2^n C_n$
 - (d) $C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots + C_{n-r} C_n = 2^n C_{n-r}$
 where $C_0, C_1, C_2, C_3, \dots$ represent ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, \dots$
- (vi) **General Binomial theorem:**
 If $n \in \mathbb{R} - 1 < x < 1$, then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)(n-2)}{r!}x^r + \dots \infty$$

(vii) **Important Expansions:**

- (a) $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r + \dots$
- (b) $(1-x)^{-1} = 1 + x^1 + x^2 + x^3 + \dots + x^r + \dots$
- (c) $(1+x)^{-2} = 1 - {}^2C_1x + {}^3C_2x^2 - {}^4C_3x^3 + \dots + (-1)^r {}^{r+1}C_r x^r + \dots$
- (d) $(1+x)^{-2} = 1 + {}^2C_1x + {}^3C_2x^2 + {}^4C_3x^3 + \dots + {}^{r+1}C_r x^r + \dots$

UNIT – 8: SEQUENCES & SERIES

1. Arithmetic Progression

(i) $T_n = a + (n-1)d, n \in \mathbb{I}$

(ii) $S_n = \frac{n}{2} [2a + (n-1)d], n \in \mathbb{I}$

where $T_n = n$ th term of an AP

$S_n =$ sum of n terms of an AP

$a =$ first term of an AP

$d =$ Common difference

(iii) Arithmetic Mean (AM) of two quantities a and b ,

$$A = \frac{a+b}{2}$$

(iv) AM of n quantities,

$$A_n = \frac{x_1 + x_2 + \dots + x_n}{n}$$

(v) $T_n = S_n - S_{n-1} (n \geq 2)$

2. Geometric Progression (GP)

(i) $T_n = ar^{n-1}$ where $a =$ first term of the GP, $r =$ common ratio of the GP and $T_n = n$ th term of the GP

(ii) Sum of n terms of a GP

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

(iii) Sum of an infinite number of terms of a GP, when $|r| < 1$

$$S_{\infty} = \frac{a}{1-r}$$

- (iv) Geometric mean of two quantities a and b ,

$$G^2 = ab, ab > 0$$

- (v) Geometric mean of n given quantities, $a_1, a_2, \dots, a_n$.

$$G_n = (a_1 a_2 a_3 \dots a_{n-1} a_n)^{1/n}$$

- (vi) Common ratio $= \frac{T_n}{T_{n-1}}$.

3. Harmonic Progression (HP)

- (i) n th term of a HP,

$$T_n = \frac{1}{a + (n-1)d}$$

- (ii) Harmonic mean of a and b ,

$$H = \frac{2ab}{a+b}$$

- (iii) Relation between A, G and H

$$\begin{aligned} AH &= \left(\frac{a+b}{2} \right) \left(\frac{2ab}{a+b} \right) \\ &= ab = G^2 \end{aligned}$$

$$\Rightarrow AH = G^2$$

- (iv) $A > G > H$, i.e., A, G, H are in descending order of magnitude.

4. Arithmetico-Geometric Sequence

Sum of an infinite number of terms of the sequence $ab, (a+d)br, (a+2d)br^2, \dots$ is

$$S = \frac{ab}{1-r} + \frac{dbr}{(1-r)^2}$$

5. Series of natural numbers

$$(i) \quad \sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$(ii) \quad \sum n^2 = 1^2 + 2^2 + 3^2 + \dots + (n-1)^2 + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(iii) \quad \sum n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

$$(iv) \quad \sum a = a + a + \dots + a (n \text{ terms}) = na$$

6. Exponential and Logarithmic series

$$(i) \quad e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$(ii) \quad \frac{e^x}{2} \quad \frac{e^{-x}}{2} \quad 1 \quad \frac{x^2}{2!} \quad \frac{x^4}{4!} \quad \dots$$

$$(iii) \quad \frac{e^x}{2} \quad \frac{e^{-x}}{2} \quad x \quad \frac{x^3}{3!} \quad \frac{x^5}{5!} \quad \dots$$

$$(iv) \quad a^x = 1 + x \ln a + \frac{x^2}{2!} (\ln a)^2 + \frac{x^3}{3!} (\ln a)^3 + \dots$$

$$(v) \quad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, |x| < 1$$

$$(vi) \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots, n \in \mathbb{R}$$

UNIT -: 9 DIFFERENTIAL CALCULUS

Functions

Formulae for domain of a function

1. Domain $(f(x) + g(x)) = \text{Domain } f(x) \cap \text{Domain } g(x)$
2. Domain $(f(x) \cdot g(x)) = \text{Domain } f(x) \cap \text{Domain } g(x)$
3. Domain $\left(\frac{f(x)}{g(x)} \right) = \text{Domain } f(x) \cap \text{Domain } g(x) \cap \{x : g(x) \neq 0\}$
4. Domain $\sqrt{f(x)} = \text{Domain } f(x) \cap \{x : f(x) \geq 0\}$
5. Domain (fog) = Domain $g(x)$, where fog is defined by $\text{fog}(x) = f(g(x))$

Odd and Even Function

- (i) A function is an odd function if $f(-x) = -f(x)$ for all x .
- (ii) A function is an even function if $f(-x) = f(x)$ for all x .

Periodic function

A function $f(x)$ is said to be a periodic function of x , if there exist a positive real number T such that $f(x + T) = f(x)$ for all x .

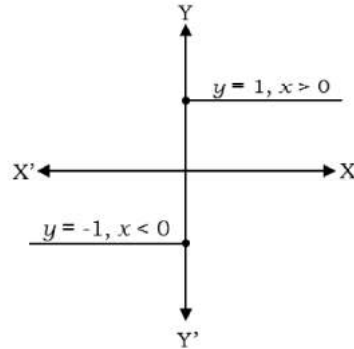
If positive value of T independent of x then $f(x)$ is periodic function and if the value of T depends upon x , then $f(x)$ is non-periodic.

Signum function

The signum function f is defined as

$$\text{Sgn } f(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

As shown in figure



Greatest Integer function

If $f(x) = [x + n]$, where $n \in \mathbb{I}$ and $[.]$ denotes the greatest integer function, then $f(x) = n + [x]$.

- a) $x - 1 < [x] \leq x$
- b) $[-x] = -[x]$, if $x \in \mathbb{I}$
- c) $[-x] = -[x] - 1$, if $x \notin \mathbb{I}$
- d) $[x] \geq n \Rightarrow x \geq n, n \in \mathbb{I}$
- e) $[x] \leq n \Rightarrow x < n + 1, n \in \mathbb{I}$
- f) $[x] > n \Rightarrow x \geq n + 1, n \in \mathbb{I}$
- g) $[x] < n \Rightarrow x < n, n \in \mathbb{I}$

Modulus function (or Absolut-Value-function)

Modulus function is given by

$$y = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Properties of Modulus Function

- a) $|x| \leq a \Rightarrow -a \leq x \leq a; (a \geq 0)$

b) $|x| \geq a \Rightarrow -a \text{ or } x \geq a; (a \geq 0)$

c) $|x \pm y| \leq |x| + |y|$

d) $|x \pm y| \geq |x| - |y|$

Limits and Continuity

Frequently used Limits

(i) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e = \lim_{h \rightarrow 0} (1+h)^{1/h}$

(ii) $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$

(iii) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$, where $n \in \mathbb{Q}$

(iv) $\lim_{\theta \rightarrow 0} \frac{\sin \theta^c}{\theta^c} = 1$ and $\lim_{\theta \rightarrow 0} \frac{\tan \theta^c}{\theta^c} = 1$

(v) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a (a > 0)$

(vi) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

(vii) $\lim_{x \rightarrow 0} \frac{\log_a (1+x)}{x} = \log_a a \quad (a > 0, a \neq 1)$

Expansions of Trigonometric Functions

(i) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

(ii) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

(iii) $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$

Sandwich Theorem

If f, g, h are functions, such that $f(x) \leq g(x) \leq h(x)$ then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x) \leq \lim_{x \rightarrow a} h(x)$$

Indeterminate Forms

If a function $f(x)$ takes any of the following forms at $x = a$

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 0^0, \infty^0, 1^\infty$$

then $f(x)$ is said to be indeterminate at $x = a$.

L'Hospital's Rule

Let $f(x)$ and $g(x)$ be two functions, such that $f(a) = 0$ and $g(a) = 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Provided $f'(a)$ and $g'(a)$ are both non-zero.

Derivatives

Differentiability and Continuity

For a function $f(x)$

- (a) Differentiable $\Rightarrow$ Continuous
- (b) Continuous $\nRightarrow$ Differentiable
- (c) Not Continuous $\Rightarrow$ Not Differentiable

Theorems on Derivatives

- (i) $\frac{d}{dx} \{f_1(x) \pm f_2(x)\} = \frac{d}{dx} f_1(x) \pm \frac{d}{dx} f_2(x)$
- (ii) $\frac{d}{dx} (k(x)) = k \frac{d}{dx} f_1(x)$, where k is any constant.
- (iii) $\frac{d}{dx} \{f_1(x) \cdot f_2(x)\} = f_1(x) \frac{d}{dx} f_2(x) + f_2(x) \frac{d}{dx} f_1(x)$
- (iv) If $y = f_1(u)$, $u = f_2(v)$ and $v = f_3(x)$ then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$

Some Standard differentials

- (i) $\frac{d}{dx} x^n = nx^{n-1}, x \in \mathbb{R}, n \in \mathbb{R}, x > 0$
- (ii) $\frac{d}{dx} (\sqrt{x}) = \frac{1}{2\sqrt{x}}$
- (iii) $\frac{d}{dx} (e^x) = e^x$
- (iv) $\frac{d}{dx} (a^x) = a^x \ln a$

$$(v) \quad \frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$(vii) \quad \frac{d}{dx} (\sin x) = \cos x$$

$$(ix) \quad \frac{d}{dx} (\tan x) = \sec^2 x$$

$$(xi) \quad \frac{d}{dx} (\sec x) = \sec x \tan x$$

$$(xiii) \quad \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$(xv) \quad \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(xvii) \quad \frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$(vi) \quad \frac{d}{dx} (\log_a x) = \frac{1}{x} \log_a e$$

$$(viii) \quad \frac{d}{dx} (\cos x) = -\sin x$$

$$(x) \quad \frac{d}{dx} (\csc x) = -\csc x \cot x$$

$$(xii) \quad \frac{d}{dx} (\operatorname{cosec} x) = \operatorname{cosec} x \cot x$$

$$(xiv) \quad \frac{d}{dx} (\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$(xvi) \quad \frac{d}{dx} (\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$(xviii) \quad \frac{d}{dx} (\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

Some Standard Substitutions

Expression	Substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$ or $a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$ or $a \cot \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$ or $a \operatorname{cosec} \theta$
$\sqrt{\frac{a+x}{a-x}}$ or $\sqrt{\frac{a-x}{a+x}}$	$x = a \cos \theta$ or $a \cos 2\theta$
$\sqrt{(2ax - x^2)}$	$x = a(1 - \cos \theta)$

Critical Points

The points on the curve $y = f(x)$ at which $\frac{dy}{dx} = 0$ or $\frac{dy}{dx}$ does not exist are known as the critical points.

Rolle's Theorem

If a function $f(x)$ is defined on $[a, b]$ satisfying

- (i) f is continuous on $[a, b]$
- (ii) f is differentiable on (a, b)
- (iii) $f(a) = f(b)$ then $c \in (a, b)$ such that $f'(c) = 0$

Lagrange's mean value Theorem

If a function $f(x)$ is defined on $[a, b]$ satisfying

- (i) f is continuous on $[a, b]$
- (ii) f is differentiable on (a, b) then $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

Test of Monotonicity

- (i) The function $f(x)$ is **monotonically increasing** in the interval $[a, b]$, if $f'(x) \geq 0$ in $[a, b]$
- (ii) The function $f(x)$ is **strictly increasing** in the interval $[a, b]$, if $f'(x) > 0$ in $[a, b]$
- (iii) The function $f(x)$ is **monotonically decreasing** in the interval $[a, b]$, if $f'(x) \leq 0$ in $[a, b]$
- (iv) The function $f(x)$ is **strictly decreasing** in the interval $[a, b]$, if $f'(x) < 0$ in $[a, b]$

Working Rule for Finding Maxima and Minima

(a) First Derivative Test

To check the maxima or minima at $x = a$

- (i) If $f'(x) > 0$ at $x < a$ and $f'(x) < 0$ at $x > a$ i.e., the sign of $f'(x)$ changes from +ve to -ve, then $f(x)$ has a local maximum at $x = a$.
- (ii) If $f'(x) < 0$ at $x < a$ and $f'(x) > 0$ at $x > a$ i.e., the sign of $f'(x)$ changes from -ve to +ve, then $f(x)$ has a local minimum at $x = a$.
- (iii) If the sign of $f'(x)$ does not change, then $f(x)$ has neither local maximum nor local minimum at $x = a$, then point ' a ' is called a point of **inflection**.

(b) Second Derivative Test

- (i) If $f''(a) < 0$ and $f'(a) = 0$, then ' a ' is a point of local maximum.
- (ii) If $f''(a) > 0$ and $f'(a) = 0$, then ' a ' is a point of local minimum.

(iii) If $f''(a) = 0$ and $f'(a) = 0$, then further differentiate and obtain $f'''(a)$

(iv) If $f'(a) = f''(a) = f'''(a) = \dots = f^{(n-1)}(a) = 0$ and $f^{(n)}(a) \neq 0$

If n is odd then $f(x)$ has neither local maximum nor local minimum at $x = a$, then point ' a ' is called a point of **inflection**.

If n is even, then if $f^{(n)}(a) < 0$ then $f(x)$ has a local maximum at $x = a$ and if $f^{(n)}(a) > 0$ then $f(x)$ has a local minimum at $x = a$.

Note: Maximum or Minimum values are also called local extremum values. For the points of local extremum either $f'(x) = 0$ or $f'(x)$ does not exist.

UNIT – 10: INTEGRAL CALCULUS

Methods of Integration

List of Basic Forms of Integrals

1. $\int f(\phi(x))\phi'(x) dx$

Substitution $\phi(x) = t$

2. $\int f(x)\phi'(x) dx$

Integration by parts

$$\int f(x)\phi'(x) dx = f(x)\phi(x) - \int \phi(x)f'(x) dx$$

3. $\int e^{ax} p_n(x) dx$, where $p_n(x)$ is polynomial of degree n .

Applying the formula for multiple integration by parts (see above), we get

$$\int e^{ax} p_n(x) dx = e^{ax} \left[\frac{p_n(x)}{a} - \frac{p_n'(x)}{a^2} + \frac{p_n''(x)}{a^3} - \dots + (-1)^n \frac{p_n^{(n)}(x)}{a^{n+1}} \right] + C$$

4. $\int \frac{Mx + N}{x^2 + px + q} dx$. $p^2 - 4q < 0$

Substitution, $x + \frac{p}{2} = t$

5. $I_n = \int \frac{dx}{(x^2 + 1)^n}$

Reduction formula is used

$$I_n = \frac{x}{(2n-2)(x^2+1)^{n-1}} + \frac{2n-3}{2n-2} I_{n-1}$$

6. $\int \frac{P(x)}{Q(x)} dx$, where $\frac{P(x)}{Q(x)}$ is a proper rational fraction

$$Q(x) = (x-x_1)^1 (x-x_2)^m \dots (x^2+px+q) \dots$$

Integrand is expressed in the form of a sum of partial fractions

$$\begin{aligned} \int \frac{P(x)}{Q(x)} = & \frac{A_1}{(x-x_1)} + \frac{A_2}{(x-x_1)^2} + \dots + \frac{A_1}{(x-x_1)^1} + \frac{B_1}{(x-x_1)} + \frac{B_2}{(x-x_1)^2} + \dots \\ & \dots + \frac{B_m}{(x-x_2)^m} + \dots + \frac{M_1x+N_1}{x^2+px+q} + \frac{M_2x+N_2}{(x^2+px+q)^2} + \dots + \frac{M_kx+N_k}{(x^2+px+q)^k} + \dots \end{aligned}$$

7. $\int \frac{Mx+N}{\sqrt{ax^2+bx+c}} dx$

By the substitution $x + \frac{b}{2a} = t$ the integral is reduced to a sum of two integrals:

$$\int \frac{Mx+N}{\sqrt{ax^2+bx+c}} dx = M_1 \int \frac{t dt}{\sqrt{at^2+m}} + N_1 \int \frac{dt}{\sqrt{at^2+m}}$$

The first integral is reduced to the integral of a power function and the second one is a tabular integral.

8. $\int R\left(x, \sqrt{ax^2+bx+c}\right) dx$, where R is a rational function of x and $\sqrt{ax^2+bx+c}$.

Reduced to an integral of rational fraction by the Euler substitutions:

$$\begin{aligned} \sqrt{ax^2+bx+c} &= t \pm x\sqrt{a} \quad (a > 0) \\ \sqrt{ax^2+bx+c} &= tx \pm \sqrt{c} \quad (c > 0) \\ \sqrt{ax^2+bx+c} &= t(x-x_1) \quad (4a-b^2 < 0) \end{aligned}$$

Where x_1 is the root of the trinomial ax^2+bx+c .

The indicated integral can also be evaluated by the trigonometric substitutions:

$$x + \frac{b}{2a} = \begin{cases} \frac{\sqrt{b^2 - 4ac}}{2a} \sin t \\ \frac{\sqrt{b^2 - 4ac}}{2a} \cot t \end{cases} \quad (a < 0, ac - b^2 < 0)$$

$$x + \frac{b}{2a} = \begin{cases} \frac{\sqrt{b^2 - 4ac}}{2a} \sec t \\ \frac{\sqrt{b^2 - 4ac}}{2a} \operatorname{cosec} t \end{cases} \quad (a > 0, 4ac - b^2 < 0)$$

$$x + \frac{b}{2a} = \begin{cases} \frac{\sqrt{4ac - b^2}}{2a} \tan t \\ \frac{\sqrt{4ac - b^2}}{2a} \cot t \end{cases} \quad (a > 0, 4ac - b^2 < 0)$$

9. $\int \frac{P_n(x)}{\sqrt{ax^2 + bx + c}} dx$, where $P_n(x)$ is a polynomial of degree n .

write the equality

$$\int \frac{P_n(x) dx}{\sqrt{ax^2 + bx + c}} = Q_{n-1}(x) \sqrt{ax^2 + bx + c} + k \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

Where $Q_{n-1}(x)$ is a polynomial of degree $n - 1$. Differentiating both parts of this equality and multiplying by $\sqrt{ax^2 + bx + c}$, we get the identity

$$P_n(x) = Q_{n-1}(x)(ax^2 + bx + c) + \frac{1}{2} Q_{n-1}(x)(2ax + b) + k,$$

Which gives a system of $n + 1$ linear equations for determining the coefficients of the polynomial $Q_{n-1}(x)$ and factor k .

And the integral $\int \frac{dx}{\sqrt{ax^2 + bx + c}}$ is taken by the method considered in

No. 10 ($M = 0$; $N = 0$)

10. $\int \frac{dx}{(x - x_1)^m \sqrt{ax^2 + bx + c}}$ 3

This integral is reduced to the above-considered integral by the substitution

$$x - x_1 = \frac{1}{t}$$

11. $\int R(\sin x, \cos x) dx$

Universal substitution $\tan \frac{x}{2} = t$

If $R(-\sin x, \cos x) = -R(\sin x, \cos x)$, then the substitution $\cos x = t$ is applied

If $R(\sin x, -\cos x) = -R(\sin x, \cos x)$, then the substitution $\sin x = t$ is applied

If $R(-\sin x, -\cos x) = -R(\sin x, \cos x)$, then the substitution $\tan x = t$ is applied

12. $\int \sin ax \sin bx dx$

$$\int \sin ax \cos bx dx$$

$$\int \cos ax \cos bx dx$$

Transform the product of trigonometric function into a sum or difference, using one of the following formulas;

$$\sin ax \sin bx = \frac{1}{2} [\cos(a-b)x - \cos(a+b)x]$$

$$\cos ax \cos bx = \frac{1}{2} [\cos(a-b)x + \cos(a+b)x]$$

$$\sin ax \cos bx = \frac{1}{2} [\sin(a-b)x + \sin(a+b)x]$$

13. $\int \sin^m x \cos^n x dx$

Where m and n are integers.

If m is an odd positive number, then apply the substitution $\cos x = t$

If n is an odd positive number, apply the substitution $\sin x = t$

If $m + n$ is an even number, apply the substitution $\tan x = t$

If m and n are even non-negative numbers, use the formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2}; \cos^2 x = \frac{1 + \cos 2x}{2}$$

14. $\int \sin^p x \cos^q x \, dx \quad (0 < x < \pi/2)$

p and q –rational numbers.

Reduce to the integral of the binomial differential by the substitution $\sin x = t$

$$\int \sin^p x \cos^q x \, dx = \int t^p (1-t^2)^{q-1} dt \quad (\text{see No. 14})$$

15. $\int R(e^{ax}) \, dx$.

Transform into an integral of a rational function by the substitution

$$e^{ax} = t$$

Indefinite Integration

1. Properties of Indefinite Integration

$$\begin{aligned} \text{(i)} \quad \int [f_1(x) \pm f_2(x) \pm \dots \pm f_n(x)] \, dx \\ = \int f_1(x) \, dx \pm \int f_2(x) \, dx \pm \dots \pm \int f_n(x) \, dx + C \end{aligned}$$

$$\text{(ii)} \quad = \int k f(x) \, dx = k \int f(x) \, dx + C$$

$$\text{(iii)} \quad \int [f(x)]^n f'(x) \, dx = \frac{[f(x)]^{n+1}}{(n+1)} + C$$

$$\text{(iv)} \quad \int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)| + C$$

$$\text{(v)} \quad \int \frac{f'(x)}{f(x)} \, dx = 2\sqrt{f(x)} + C$$

2. Standard Results in Integration

$$\text{(i)} \quad \int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\text{In particular } \int 1 \, dx = \int x^0 \, dx = x + C$$

$$\text{(ii)} \quad \int \frac{1}{x} \, dx = \ln |x| + C$$

$$(iii) \quad \int e^x dx = e^x + C$$

$$(iv) \quad \int a^x dx = \frac{a^x}{\ln a} + C; a \neq 1, a > 0$$

$$(v) \quad \int \sin x dx = -\cos x + C$$

$$(vi) \quad \int \cos x dx = \sin x + C$$

$$(vii) \quad \int \tan x dx = \ln |\sec x| + C$$

$$(viii) \quad \int \cot x dx = \ln |\sin x| + C$$

$$(ix) \quad \int \sec x dx = \ln |\sec x + \tan x| + C = -\ln |\sec x - \tan x| + C \\ = \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C$$

$$(x) \quad \int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x - \cot x| + C = \ln |\operatorname{cosec} x + \cot x| + C \\ = \ln \left| \tan \frac{x}{2} \right| + C$$

$$(xi) \quad \int \sec^2 x dx = \tan x + C$$

$$(xii) \quad \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$(xiii) \quad \int \sec x \tan x dx = \sec x + C$$

$$(xiv) \quad \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$(xv) \quad \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$(xvi) \quad \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$(xvii) \quad \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$(xviii) \int \frac{dx}{\sqrt{(a^2 - x^2)}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$(xix) \int \frac{dx}{\sqrt{(x^2 - a^2)}} = \ln \left| x + \sqrt{(x^2 - a^2)} \right| + C$$

$$(xx) \int \frac{dx}{\sqrt{(x^2 + a^2)}} = \ln \left| x + \sqrt{(x^2 + a^2)} \right| + C$$

$$(xxi) \int \frac{dx}{x\sqrt{(x^2 - a^2)}} = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$$

$$(xxii) \int \sqrt{(a^2 - x^2)} dx = \frac{x}{2} \sqrt{(a^2 - x^2)} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$(xxiii) \int \sqrt{(a^2 + x^2)} dx = \frac{x}{2} \sqrt{(a^2 + x^2)} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$(xxiv) \int \sqrt{(x^2 - a^2)} dx = \frac{x}{2} \sqrt{(x^2 - a^2)} - \frac{a^2}{2} \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$(xxv) \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$$

$$(xxvi) \int \frac{1}{x^n} dx = \frac{-1}{(n-1)x^{n-1}} + C$$

Definite Integration

Properties of Definite Integrals

$$(i) \int_a^b f(x) dx = \int_a^b f(t) dt$$

$$(ii) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$(iii) \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Where c , is a point within or out of the interval $[a, b]$.

$$(iv) \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$(v) \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ or } 0$$

According as $f(x)$ is even or odd function of x .

$$(vi) \int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

$$(vii) \int_a^b f(x) dx = (b-a) \int_0^1 f((b-a)x+a) dx$$

$$(viii) \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$(ix) \int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx$$

where $f(x)$ is periodic with period T and $n \in \mathbb{I}$.

$$(x) \text{ If } f(a+x) = f(x), \text{ then } \int_0^{na} f(x) dx = n \int_0^a f(x) dx$$

$$(xi) \text{ If } f(x) \leq \phi \text{ for } x \in [a, b], \text{ then } \int_a^b f(x) dx \leq \int_a^b \phi(x) dx$$

$$(xii) \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

(xiii) **Leibnitz's rule:**

If f continuous on $[a, b]$ and $u(x)$ and $v(x)$ are differentiable functions of x whose values lie in $[a, b]$ then

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f\{v(x)\} \cdot \frac{dv}{dx} - f\{u(x)\} \cdot \frac{du}{dx}$$

UNIT - 11: DIFFERENTIAL EQUATIONS

(i) $d(xy) = x dy + y dx$

(ii) $d\left(\frac{x}{y}\right) = \frac{y dx - x dy}{y^2}$

(iii) $d\left(\tan^{-1} \frac{x}{y}\right) = \frac{y dx - x dy}{x^2 + y^2}$

(iv) $d[\ln(xy)] = \frac{x dy - y dx}{xy}$

(v) $d\left(\ln\left(\frac{x}{y}\right)\right) = \frac{y dx - x dy}{xy}$

(vi) $d\left[\frac{1}{2} \ln(x^2 + y^2)\right] = \frac{x dx + y dy}{x^2 + y^2}$

(vii) $d(x^p y^q) = x^{p-1} y^{q-1} (py dx + qx dy)$

UNITS - 17: TRIGONOMETRY

Trigonometric Functions and Identities

1. Measurement of an Angle

(i) $1^\circ = 60 \text{ minutes} = 60'$

(ii) $1' = 60 \text{ seconds} = 60''$

(iii) Each interior angle of a regular polygon of n sides is equal to $\frac{(n-2)\pi}{n}, n > 2$

2. Some Important Formulae and Identities

$$\sin(A+B) = \sin A \cos B + \cos A \sin B \quad \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B \quad \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}; \quad \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\tan(45^\circ + A) = \frac{1 + \tan A}{1 - \tan A}; \quad \tan(45^\circ - A) = \frac{1 - \tan A}{1 + \tan A}$$

$$\sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$\cos(A+B) \cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B) \quad 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B) \quad 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2} \quad \sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2} \quad \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2}$$

$$\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A} \quad \cos 2A = \begin{cases} 1 - \sin^2 A \\ \cos^2 A - \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A} \\ 2 \cos^2 A - 1 \end{cases}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} \quad \sin^2 A = \frac{1 - \cos 2A}{2} \text{ and } \cos^2 A = \frac{1 + \cos 2A}{2}$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A; \quad \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

3. Conditional Identities

If $A + B + C = 180^\circ$, then

$$(i) \quad \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$(ii) \quad \cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$$

$$(iii) \quad \sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$(iv) \quad \cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$(v) \quad \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$(vi) \quad \tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$$

4. Trigonometrical Ratios

$$(i) \quad \sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$$

$$(ii) \quad \sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$$

$$(iii) \quad \sin^2 \theta + \cos^4 \theta = \cos^2 \theta + \sin^4 \theta = 1 - \sin^2 \theta \cos^2 \theta$$

$$(iv) \quad \sin^2 \theta + \operatorname{cosec}^2 \theta \geq 2, \cos^2 \theta + \sec^2 \theta \geq 2 \quad \sec^2 \theta + \operatorname{cosec}^2 \theta \geq 4$$

5. Some standard Results

$$(i) \quad \tan \theta + \cot \theta = 2 \operatorname{cosec} 2\theta$$

$$(ii) \quad \cot \theta - \tan \theta = 2 \cot 2\theta$$

$$(iii) \quad \sin \theta + \cos \theta \text{ has the same sign as that of } \sin \left(\theta + \frac{\pi}{4} \right)$$

$$(iv) \quad \sin \theta - \cos \theta \text{ has the same sign as that of } \sin \left(\theta - \frac{\pi}{4} \right)$$

$$(v) \quad \text{Maximum value of } a \sin x + b \cos x \text{ is } \sqrt{a^2 + b^2} \text{ and its minimum value is } -\sqrt{a^2 + b^2}.$$

$$(vi) \quad \text{The equation } a \sin x + b \cos x = c \text{ has real solutions only if } |c| \leq \sqrt{a^2 + b^2}, \text{ i.e., if } c^2 \leq a^2 + b^2$$

$$(vii) \quad \tan (x_1 + x_2 + x_3 + \dots + x_n) = \frac{S_1 - S_3 + S_5 - \dots}{1 - S_2 + S_4 - \dots},$$

UNIT – 12 : TWO DIMENSIONAL GEOMETRY

Straight Lines

1. Distance Formula

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Note : Distance of (x_1, y_1) from origin $= \sqrt{x_1^2 + y_1^2}$

2. Section formula

If $R(x, y)$ divides the line joining $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio $m_1 : m_2$ ($m_1, m_2 > 0$) then

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}; \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \quad (\text{divides internally})$$

$$\text{and } x = \frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}; \quad y = \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \quad (\text{divides externally})$$

The mid point of PQ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$.

3. Area of a Triangle

The area of a ΔABC with vertices $A(x_1, y_1)$; $B(x_2, y_2)$ and $C(x_3, y_3)$ is denoted by Δ and is given as:

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

4. Standard points of a Triangle

- (i) The centroid of a triangle with vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

- (ii) The incentre of the triangle ABC is

$$\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$$

a, b, c being the sides BC, CA, AB of the triangle respectively.

- (iii) The orthocenter of the triangle ABC is

$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

- (iv) The circumcentre of the triangle ABC is

$$\left(\frac{x_1 \tan 2A + x_2 \tan 2B + x_3 \tan 2C}{\tan 2A + \tan 2B + \tan 2C}, \frac{y_1 \tan 2A + y_2 \tan 2B + y_3 \tan 2C}{\tan 2A + \tan 2B + \tan 2C} \right)$$

5. Collinearity of three Given Points

The three given points are collinear i.e., lie on the same straight line if

- (i) Area of triangle ABC is zero.
- (ii) Slope of AB = slope of BC = slope of AC
- (iii) Distance between A and B + distance between B and C = Distance between A and C
- (iv) Find the equation of the line passing through any two points, if third point satisfied the equation of the line then three points are collinear.

6. Slope of a line

The slope of a line joining two points (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta \quad (x_1 \neq x_2)$$

where θ is angle which the line makes with the positive direction of x -axis.

7. Equation of Straight line in Various Forms

- (i) **General Form** : The general equation of the first degree in x and y is $ax + by + c = 0$, where a and b can not be zero at the same time.

$$\text{Its slope is } = -\frac{a}{b}$$

$$\text{Intercept on the } x\text{-axis is } = -\frac{c}{a}$$

$$\text{Intercept on the } y\text{-axis } = -\frac{c}{b}$$

- (ii) **Intercepts form** : The equation of a line making intercepts ' a ' and ' b ' upon x and y axes respectively is given by

$$\frac{x}{a} + \frac{y}{b} = 1$$

- (iii) **Slope-intercept form** : The equation of line which has slope m and cuts off an intercept c upon y -axis is given by $y = mx + c$ where $m = \tan \theta$

- (iv) **Point-Slope form** : The equation of a line passing through the point (x_1, y_1) and having slope m is given by

$$y - y_1 = m(x - x_1)$$

- (v) **Two point Form** : The equation of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by $(y - y_1) = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$

- (vi) **Parametric Form** : The equation of a line passing through x_1, y_1 and making an angle θ with the positive direction of x -axis

$$\text{is } \frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$$

Where r is the distance of the point (x, y) from the point (x_1, y_1) . If r is positive, then the point (x, y) is on the right of (x_1, y_1) but if r is negative then (x, y) is on the left of (x_1, y_1) .

- (vii) **Normal Form** : The equation of a line on which the perpendicular from origin is of length p and the perpendicular makes an angle α with the positive direction of x -axis is given by, $x \cos \alpha + y \sin \alpha = p$

8. Positions of points (x_1, y_1) and (x_2, y_2) relative to a given line

If the points (x_1, y_1) and (x_2, y_2) are on the same side of the line

$ax + by + c = 0$, then $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ both are of the same sign and hence $\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} > 0$, and if the points (x_1, y_1) and (x_2, y_2) are

on the opposite of the line $ax + by + c = 0$, then $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ both are of signs opposite to each other and hence

$$\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} < 0$$

9. Angle between two Lines

Angle between two lines whose slopes are m_1 and m_2 is $\theta = \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Corollary 1 : If two lines whose slopes are m_1 and m_2 are parallel if

$$\theta = 0 \text{ (or } \pi) \Leftrightarrow \tan \theta = 0 \Leftrightarrow m_1 = m_2.$$

Corollary 2 : If two lines whose slopes are m_1 and m_2 are perpendicular

$$\text{if } \theta = \frac{\pi}{2} \left(\text{or } -\frac{\pi}{2} \right)$$

$$\Rightarrow \cot \theta = 0 \Rightarrow m_1 \cdot m_2 = -1.$$

Note : Two lines given by the equations $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$ are

(i) Parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

(ii) Perpendicular if $a_1 a_2 + b_1 b_2 = 0$

(iii) Identical if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

10. Length of perpendicular from a Point on a Line

The length of perpendicular from (x_1, y_1) on $ax + by + c = 0$ is

$$\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

11. Distance between Two Parallel Line.

The perpendicular distance between the parallel lines $ax + by + c = 0$ and $ax + by + c_1 = 0$ is

$$\left| \frac{c_1 - c}{\sqrt{a^2 + b^2}} \right|$$

12. Family of Lines

Any line passing through the point of intersection of the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ can be represented by the equation $(a_1x + b_1y + c_1) + \lambda(a_2x + b_2y + c_2) = 0$.

13. Concurrent Lines

The three lines $a_ix + b_iy + c_i = 0, i=1,2,3$ are concurrent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

14. Equations of the bisectors of the angles between two Lines

Equations of the bisectors of the lines

$$L_1 : a_1x + b_1y + c_1 = 0 \text{ and } L_2 : a_2x + b_2y + c_2 = 0$$

$$(a_1b_2 \neq a_2b_1) \text{ where } c_1 > 0 \text{ and } c_2 > 0 \text{ are}$$

$$\frac{(a_1x + b_1y + c_1)}{\sqrt{(a_1^2 + b_1^2)}} = \pm \frac{(a_2x + b_2y + c_2)}{\sqrt{(a_2^2 + b_2^2)}}$$

Conditions	Obtuse angle bisector	Acute angle bisector
$a_1a_2 + b_1b_2 < 0$	-	+
$a_1a_2 + b_1b_2 > 0$	+	-

Pair of Straight Lines

1. Homogeneous Equation of Second Degree

An equation of the form $ax^2 + 2hxy + by^2 = 0$ is called a homogeneous equation of second degree. It represents two straight lines through the origin.

- (i) The lines are real and distinct if $h^2 - ab > 0$.
- (ii) The lines are coincident if $h^2 - ab = 0$.
- (iii) The lines are imaginary if $h^2 - ab < 0$.
- (iv) If the lines represented by $ax^2 + 2hxy + by^2 = 0$ are $y - m_1x = 0$ and $y - m_2x = 0$ then $(y - m_1x)(y - m_2x) = y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 = 0$
 $\Rightarrow m_1 + m_2 = -\frac{2h}{b}, m_1m_2 = \frac{a}{b}$

2. Angle between two Lines

If $a + b \neq 0$ and θ is the actual angle between the lines $ax^2 + 2hxy + by^2 = 0$, then

$$\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|$$

Note: Lines $ax^2 + 2hxy + by^2 = 0$ are mutually perpendicular iff $a + b = 0$

3. Equation of the Bisectors of the Angles between the Lines
 $ax^2 + 2hxy + by^2 = 0$

The equation of bisectors is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

4. General equation of Second Degree

The equation, $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, is the general second degree equation.

It represents a pair of straight lines if

$$\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

i.e., if $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

otherwise it represents a conic (i.e., if $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$)

Circle and Family of circles

1. The equation of a circle with centre (h, k) and radius r is $(x - h)^2 + (y - k)^2 = r^2$. If the centre is at the origin, the equation of circle is $x^2 + y^2 = r^2$.
2. Equation of the circle on the line segment joining (x_1, y_1) and (x_2, y_2) as diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$
3. The general equation of a circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Where g, f, c are constants. The centre is $(-g, -f)$ and the radius is $\sqrt{g^2 + f^2 - c}$, ($g^2 + f^2 \geq c$).

Note : A general equation of second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

In x, y represents a circle if

- (i) Coefficient of x^2 = coefficient of y^2 i.e., $a = b \neq 0$

(ii) Coefficient of xy is zero, i.e. $h = 0$.

4. The equation of the circle through three non-collinear points A (x_1, y_1), B (x_2, y_2), C (x_3, y_3) is

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0$$

5. The point P (x_1, y_1) lies outside, on or inside the circle

$$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0. \text{ according as}$$

$$S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c > \text{ or } < 0.$$

6. The parametric co-ordinates of any points on the circle

$$(x - h)^2 + (y - k)^2 = r^2 \text{ are given by}$$

$$(h + r \cos \theta, k + r \sin \theta), (0 \leq \theta < 2\pi)$$

7. The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at the point (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$ and that of the normal is

$$y - y_1 = \frac{y_1 + f}{x_1 + g}(x - x_1)$$

The equation of tangent to the circle $x^2 + y^2 = r^2$ at the point (x_1, y_1) is

$$xx_1 + yy_1 = r^2 \text{ and that of the normal } \frac{x}{x_1} = \frac{y}{y_1}.$$

8. The general equation of a line with slope m and which is tangent to a circle

$$x^2 + y^2 = a^2 \quad y = mx \pm a\sqrt{1 + m^2}.$$

9. The locus of point of intersection of two perpendicular tangents is called the director circle. The director circle of the circle

$$x^2 + y^2 = a^2 \text{ is } x^2 + y^2 = 2a^2.$$

- 10.** Equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$
in terms of its middle point (x_1, y_1) is

$$T = S_1$$

Where

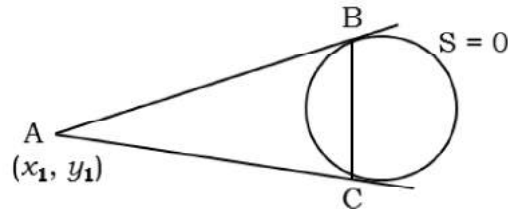
$$T = xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c; S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$

11. Equation of the Chord of contact

Equation of the chord of contact of the circle

$$xx_1 + yy_1 + 2gx + 2fy + c = 0$$

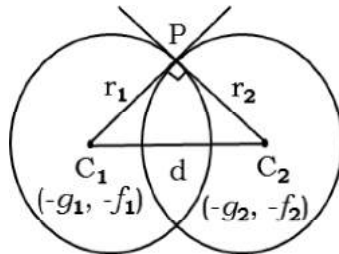
$$\text{is } xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$



- 12. Length of tangent** $= \sqrt{(x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c)} = \sqrt{S_1}$

13. Condition of orthogonality of two circles

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2 \quad \text{OR} \quad r_1^2 + r_2^2 = d^2$$



14. Pair of tangents

Tangents are drawn from $P(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$

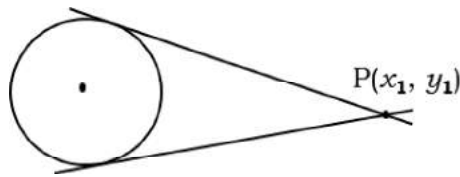
Then equation of pair of tangents is

$$SS_1 = T^2$$

Where $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$

$$S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$$

$$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$



15. **Radical Axis :** The equation of radical-axis if two circles $S_1 = 0$ and $S_2 = 0$ is given by $S_1 - S_2 = 0$ (coefficient of x^2, y^2 in S_1 and S_2 are 1).

16. Family of Circles

Let $S \equiv x^2 + y^2 + 2gx + 2fy + C = 0$

$$S' \equiv x^2 + y^2 + 2g'x + 2f'y + C' = 0$$

and $L = px + qy + r = 0$, then

- (i) If $S = 0$ and $S' = 0$ intersect in real and distinct points,
 $S + \lambda S' = 0$ ($\lambda \neq -1$) represents a family of circles passing through these points. $S - S' = 0$ (for $\lambda = -1$) represents the common chord of the circles $S = 0$ and $S' = 0$.

Conic Section

Conic is the locus of a point moving in a plane so that the ratio of its distance from a fixed point (known as focus) to its distance from a fixed line (known as directrix) is constant. This ratio is known as Eccentricity and is denoted by e .

If $e = 1$, then locus is a **Parabola**.

If $e < 1$, then locus is an **Ellipse**.

If $e > 1$, then locus is a **Hyperbola**.

Nature of Conics

The equation of conics is represented by the general equation of second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, \text{ when its discriminant, } \Delta \neq 0.$$

The discriminant of the above equation is given by

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

The following table shows the nature of conic for different condition on a , b and h .

Condition	Nature of Conic
$h = 0, a = b$	a Circle
$ab - h^2 = 0$	a Parabola
$ab - h^2 > 0$	an Ellipse
$ab - h^2 < 0$	a Hyperbola
$ab - h^2$ and $a + b = 0$	a Rectangular hyperbola.

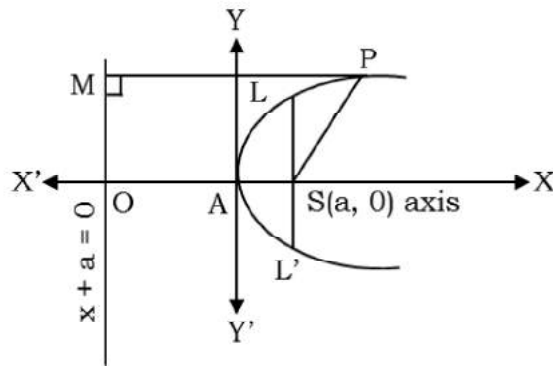
Parabola

1. Standard form of a Parabola

The general form of standard parabola is $y^2 = 4ax$, where a is constant.

2. Important Properties

- (i) $SP = PM$ and $AS = AO$
- (ii) Vertex is at origin $A \equiv (0,0)$
- (iii) Focus is at $S \equiv (a,0)$
- (iv) Directrix is $x + a = 0$
- (v) Axis is $y = 0$ (x-axis)
- (vi) Length of latus rectum = $LL' = 4a$
- (vii) Ends of the latus rectum are $L \equiv (a, 2a)$ and $L' \equiv (a, -2a)$.
- (viii) The parametric equation is : $x = at^2, y = 2at$.



3. General equation of a parabola

$$(x - a)^2 + (y - b)^2 = \frac{(lx + my + n)^2}{l^2 + m^2}$$

This equation is of the form

$$(mx - ly)^2 + 2gx + 2fy + C = 0$$

Which is the general equation of a parabola.

4. Position of a Point (h, k) with respect to a Parabola $y^2 = 4ax$

Let P be any point (h, k) .

Now P will lie outside, on or inside the parabola according as

$$(k^2 - 4ah) \geq 0.$$

5. Equation of the tangent

Equation of tangents in terms of slope, $y = mx + \frac{a}{m}, m \neq 0$.

The equation of the tangent at any point (x_1, y_1) on the parabola $y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$

Corollary 1 : Equation of tangent at any point having parameter ' t ' is

$$ty = x + at^2 \text{ Slope of tangent is } \frac{1}{t}$$

Corollary 2 : Co - ordinates of the point of intersection of tangents at ' t_1 '

$$\text{and } 't_2' \text{ is } \{at_1t_2, a(t_1 + t_2)\}.$$

Corollary 3 : If the chord joining ' t_1 ' and ' t_2 ' is a focal chord, then

$$t_1 t_2 = -1.$$

$$\Rightarrow t_2 = \frac{-1}{t_1}$$

Hence if one extremity of a focal chord is $(at_1^2, 2at_1)$, then the other extremity $(at_2^2, 2at_2)$ becomes $\left(\frac{a}{t_1^2}, -\frac{2a}{t_1}\right)$.

6. Equation of the Normal

Equation of normal at any point ' t ' is

$$y = -tx + 2at + at^3$$

Slope of normal is $-t$.

7. Equation of the Normal in terms of slope

$$y = mx - 2am - am^3$$

at the point $(am^2, -2am)$

Hence any line $y = mx + c$ will be a normal to the parabola if $c = -2am - am^3$.

8. Equation of chord with mid point (x_1, y_1)

The Equation of the chord of parabola $y^2 = 4ax$, whose mid point be (x_1, y_1) is

$$T = S_1$$

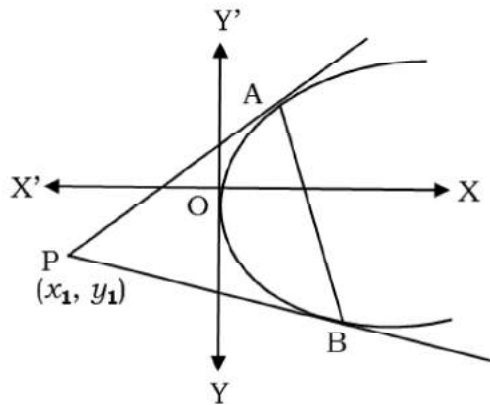
Where $T \equiv yy_1 - 2a(x + x_1) = 0$

and $S_1 = y_1^2 - 4ax_1 = 0$

9. Chord of Contact

If PA and PB be the tangents through point $P(x_1, y_1)$ (see Figure) to the parabola $y^2 = 4ax$, then the equation of the chord of contact AB is

$$yy_1 = 2a(x + x_1) \text{ or } T = 0 \text{ at } (x_1, y_1)$$



10. Pair of Tangents

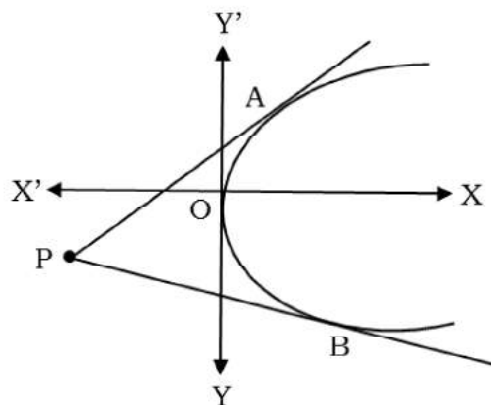
If $P(x_1, y_1)$ be any point lies outside the parabola $y^2 = 4ax$, and a pair of tangents PA, PB can be draw to it from P, (see Figure) then the equation of pair of tangents PA and PB is

$$SS_1 = T^2$$

Where, $S \equiv y^2 - 4ax = 0$

$$S_1 \equiv y_1^2 - 4ax_1 = 0$$

$$T \equiv yy_1 - 2a(x + x_1) = 0$$

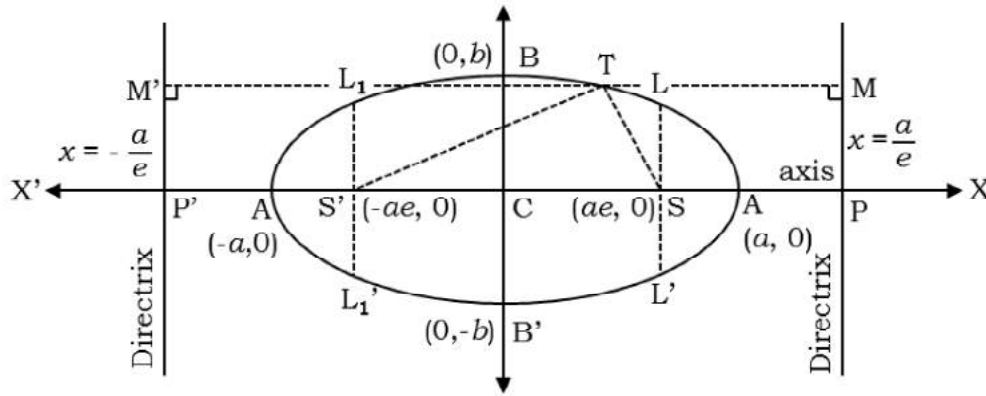


Ellipse

1. Standard form of an Ellipse

The general form of standard ellipse is :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b), \text{ where } a \text{ and } b \text{ are constants. (see Figure below)}$$



2. Important Properties

- (i) $ST = e TM$
- (ii) Co-ordinate of centre C (0,0)
- (iii) Equation of directrix $x = \pm a / e$
- (iv) Equation of latus rectum $x = \pm ae$

3. General Equation of an Ellipse

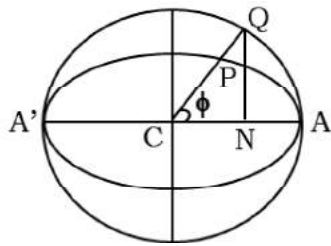
$$(l^2 + m^2) \left\{ (x-a)^2 + (y-b)^2 \right\} = e^2 (lx + my + n)^2$$

4. Parametric Equation of an Ellipse

The parametric equations of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are $x = a \cos \phi, y = b \sin \phi$, where ϕ is the parameter.

5. Auxiliary Circle and Eccentric angle

The circle described on the major axis of an ellipse as diameter is called its auxiliary circle (see Figure) .



The equation of the auxiliary circle is

$$x^2 + y^2 = a^2$$

$$\therefore Q \equiv (a \cos \phi, a \sin \phi) \text{ and } P \equiv (a \cos \phi, b \sin \phi)$$

($\phi = \text{eccentric angle}$)

6. Condition for tangency

A line $y = mx + c$ is tangent to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $c = \pm \sqrt{a^2 m^2 + b^2}$.

Corollary 1 : $x \cos \alpha + y \sin \alpha = p$ is a tangent if

$$p^2 = a^2 \cos^2 \alpha + b^2 \sin^2 \alpha.$$

Corollary 2 : $lx + my + n = 0$ is a tangent if $n^2 = a^2 l^2 + b^2 m^2$.

7. Equation of the Tangent

(i) The equation of the tangent at any point (x_1, y_1) on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1.$$

(ii) The equation of tangent at any point ' ϕ ' is $a \cos \phi + \frac{y}{b} \sin \phi = 1$.

8. Equation of the Normal

(i) The equation of the normal at any point (x_1, y_1) on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \frac{a x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(ii) The equation of the normal at any point ' ϕ ' is

$$ax \sec \phi - by \operatorname{cosec} \phi = a^2 - b^2$$

9. Equation of chord with mid point (x_1, y_1)

The equation of the chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose mid point be (x_1, y_1) is $T = S_1$

Where $T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = 0$; $S_1 \equiv \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 = 0$

10. Chord of Contact

If PA and PB be the tangents through point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then the equation of the chord of contact AB is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.

11. Pair of tangents

Let $P(x_1, y_1)$ be any point lying outside the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and a pair of tangents PA, PB can be drawn to it from P. then the equation of pair of tangents of PA and PB is

$$SS_1 = T^2$$

12. Important properties of an Ellipse

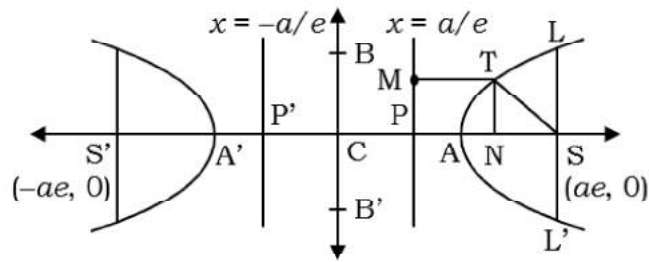
- (i) If $\alpha, \beta, \gamma, \delta$ be the eccentric angles of the four concyclic points on an ellipse then $\alpha + \beta + \gamma + \delta = 2n\pi, n \in I$
- (ii) The necessary and sufficient condition for the normal at three α, β, γ points on the ellipse to be concurrent is

$$\sin(\beta + \gamma) + \sin(\gamma + \alpha) + \sin(\alpha + \beta) = 0$$

Hyperbola**1. Standard form of a Hyperbola**

If (a, b) is the focus S, and $lx + my + n = 0$ is the equation of directrix, then the standard equation of a hyperbola is

$$(l^2 + m^2) \left\{ (x - a)^2 + (y - b)^2 \right\} = e^2 (lx + my + n)^2$$



2. Important Properties

- (i) $ST = e TM$
- (ii) Co-ordinates of vertices A and A' are $(\pm a, 0)$.
- (iii) Co-ordinates of the foci S and S' are $(\pm ae, 0)$.
- (iv) Equation of directrix $x = \pm a / e$.
- (v) Equation of latus rectum $x = \pm ae$ and length $LL' = L_1L_1 = \frac{2b^2}{a}$.

3. Parametric Equation of the Hyperbola

The parametric equations of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $x = a \sec \phi$, $y = b \tan \phi$, where ϕ is the parameter.

4. Condition for Tangency

A line $y = mx + c$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ iff $c^2 = a^2m^2 - b^2$.

5. Equation of the Tangent

- (i) The equation of the tangent at any point (x_1, y_1) on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

- (ii) The equation of tangent at any point ' ϕ ' is $a \phi - \frac{y}{b} \phi =$

6. Equation of Chord with Midpoint (x_1, y_1)

The equation of the chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose mid point be (x_1, y_1) is $T = S_1$

$$\text{Where } T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 = 0, S_1 \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 = 0$$

7. Chord of Contact

If two tangents are drawn through a point (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then the equation of the chord of contact is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \text{ or } T = 0 \text{ at } (x_1, y_1)$$

8. Pair of Tangents

If a pair of tangents is drawn from any point (x_1, y_1) outside the hyperbola, then the equation of pair of tangent is

$$SS_1 = T^2$$

$$\text{Where } S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0; S_1 \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 = 0; T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 = 0$$

9. Asymptotes of Hyperbola

A hyperbola has two asymptotes passing through its centre. Asymptotes of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are given by $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.

Note : Angle between asymptotes $= 2 \tan^{-1} \left(\frac{b}{a} \right)$.

10. Rectangular Hyperbola

If asymptotes of the standard hyperbola are perpendicular to each other it is known as rectangular hyperbola.

Properties of Rectangular Hyperbola $xy = c^2$

- (i) Eccentricity, $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{2}$.
- (ii) Since $x = ct, y = \frac{c}{t}$ satisfies $xy = c^2, (x, y) = \left(ct, \frac{c}{t}\right)$ is called a 't' point with parameter t .
- (iii) Equation of the chord joining t_1 and t_2 is
- $$x + y t_1 t_2 - c(t_1 + t_2) = 0$$
- (iv) Equation of tangent at 't' is
- $$x + y t^2 - 2ct = 0$$
- (v) Equation of normal at 't' is
- $$x t^2 - y t - c t^4 + c = 0$$
- (vi) Equation of tangent at (x_1, y_1) is
- $$x y_1 + y x_1 = 2c^2$$
- (vii) Equation of normal at (x_1, y_1) is
- $$x x_1 - y y_1 = x_1^2 - y_1^2$$

UNIT - 13 : THREE DIMENSIONAL GEOMETRY

1. **Distance formula:** The distance between two points $P(x_1, y_1, z_1)$ and

$Q(x_2, y_2, z_2)$ in space is given by $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

2. **Section formula:** If $R(x, y, z)$ divides the join of $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio $m_1 : m_2$ ($m_1, m_2 > 0$) then

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}; y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}; z = \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2}$$

(divides internally), and

$$x = \frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}; y = \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}; z = \frac{m_1 z_2 - m_2 z_1}{m_1 - m_2} \text{ (divides externally)}$$

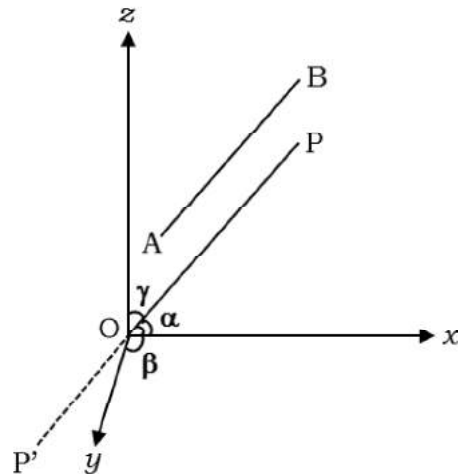
3. **Centroid of a triangle:** The centroid of a triangle ABC whose vertices are $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ are

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

4. **Direction Cosines (d.c.'s):**

If $\cos \alpha$, $\cos \beta$, $\cos \gamma$ are direction cosines of a given line AB, then

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$



5. **Direction ratios (d.r.'s):** direction ratios of a line are numbers which are proportional to the direction cosines of a line. Direction ratios of a line PQ, where P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively, are $x_2 - x_1$, $y_2 - y_1$ and $z_2 - z_1$.
6. **Relation between the d.c.'s and d.r.'s:** If a , b , c are the d.r.'s and l , m , n are the d.c.'s then

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\cos \theta = \pm \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}}$$

$$\text{and } \sin \theta = \pm \frac{\sqrt{\Sigma (b_1 c_2 - b_2 c_1)^2}}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}}$$

8. Angle between two Planes

If θ be the angle between the planes $a_1 x + b_1 y + c_1 z + d_1 = 0$ and $a_2 x + b_2 y + c_2 z + d_2 = 0$ then

$$\theta = \cos^{-1} \left(\frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}} \right)$$

Also, 1. If planes are perpendicular then $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

2. If planes are parallel then, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

9. Length of Perpendicular from a point to a Plane

The length of perpendicular from (x_1, y_1, z_1) on $ax + by + cz + d = 0$ is

$$\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{(a^2 + b^2 + c^2)}}$$

10. Family of Planes

Any plane passing through the line of intersection of the planes $ax + by + cz + d = 0$ and $a_1 x + b_1 y + c_1 z + d_1 = 0$ can be represented by the equation

$$(ax + by + cz + d) + \lambda (a_1 x + b_1 y + c_1 z + d_1) = 0$$

UNITS – 14 : VECTOR ALGEBRA

1. Position Vector of R($\bar{r}$) dividing $\overline{PQ}$ in the ratio $m : n$ is

$$\bar{r} = \frac{m\bar{q} + n\bar{p}}{m + n} \text{ [internal division] and}$$

$$\bar{r} = \frac{m\bar{q} - n\bar{p}}{m - n} \text{ [external division]}$$

2. If $\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$,

$$\bar{b} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$$

$$\bar{a} \cdot \bar{b} = |\bar{a}||\bar{b}|\cos\theta; \text{ also } \bar{a} \cdot \bar{b} = a_1b_1 + a_2b_2 + a_3b_3$$

$$\cos\theta = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}||\bar{b}|} = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

3. $\bar{a} \cdot \bar{b} = 0 \Rightarrow a = 0 \text{ or } b = 0 \text{ or } a \perp b$

4. $\bar{a} \times \bar{b} = |\bar{a}||\bar{b}|\sin\theta\bar{e}; |\bar{a} \times \bar{b}| = |\bar{a}||\bar{b}|\sin\theta$

$$\bar{a} \times \bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

5. $\bar{a} \times (\bar{b} \times \bar{c}) = (\bar{a} \cdot \bar{c})\bar{b} - (\bar{a} \cdot \bar{b})\bar{c}$

6. $(\bar{a} \times \bar{b}) \cdot \bar{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \bar{a} \cdot (\bar{b} \times \bar{c}) = [\bar{a} \bar{b} \bar{c}]$

7. $\bar{a}, \bar{b}, \bar{c}$ are coplanar if and only if $\bar{a} \cdot (\bar{b} \times \bar{c}) = 0$.

8. Area of parallelogram OACB = $|\bar{a} \times \bar{b}|$

$$\text{Area of } \Delta OAB = \frac{1}{2} |\bar{a} \times \bar{b}|$$

9. Volume of parallelepiped = $[\bar{a} \bar{b} \bar{c}]$
10. For any 3 vectors $\bar{a}, \bar{b}, \bar{c}$, $[\bar{a} \bar{b} \bar{c}] = [\bar{b} \bar{c} \bar{a}] = [\bar{c} \bar{a} \bar{b}]$
11. Unit vector $\perp \bar{a}$ and $\bar{b} = \pm \frac{\bar{a} \times \bar{b}}{|\bar{a} \times \bar{b}|}$ and $\sin \theta = \frac{|\bar{a} \times \bar{b}|}{|\bar{a}| |\bar{b}|}$

UNITS – 15 : MEASURES OF CENTRAL TENDENCY & DISPERSION

1. Arithmetic Mean (AM)

- (i) Individual Series $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$
- (ii) Discrete Series $\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{1}{N} \sum_{i=1}^n f_i x_i$

$$\text{where } N = \sum_{i=1}^n f_i$$

2. Geometric Mean (GM)

- (i) Individual series
- $$G = (x_1, x_2, \dots, x_n)^{1/n}$$
- (ii) Discrete series

$$G = (x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{1/N}, \text{ where } N = \sum_{i=1}^n f_i$$

3. Harmonic Mean (HM)

$$H = \frac{1}{\frac{1}{n} \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right)}$$

or $\frac{1}{H} = \frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}$ (Individual series)

and $\frac{1}{H} = \frac{1}{N} \sum_{i=1}^n \frac{f_i}{x_i}$ (Discrete series)

If $x_1, x_2, \dots, x_n > 0$, then it is known that

$$AM \geq GM \geq HM$$

4. Median

$$M = l + \frac{h}{f} \left(\frac{N}{2} - C \right) \text{ where, } N = \sum_{i=1}^n f_i$$

h = the width of the median class

C = the cumulative frequency (c.f.) of the preceding to the median class.

f = the frequency of the median class

l = the lower limit of the median class

5. Mode

$$\text{Mode (for continuous series)} = l + \frac{(f_1 - f_0)}{(2f_1 - f_2 - f_0)} \times h$$

where l = the lower limit of the modal class (the class having maximum frequency)

f_1 = frequency of the modal class

f_0 = frequency of the class preceding the modal class

f_2 = frequency of the class succeeding the modal class

h = width of the modal class

6. Mean - Deviation (MD)

$$MD = \frac{1}{N} \sum_{i=1}^n f_i |x_i - A|, \text{ where } N = \sum_{i=1}^n f_i$$

7. Standard Deviation (SD)

$$\alpha^2 \text{ (Variance)} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

and Standard deviation = $\sqrt{\text{Variance}}$

UNITS – 16 : PROBABILITY

1. Definition

The probability of occurrence of an event is the ratio of the number of cases in its favour to the total number of cases (equally likely).

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{number of Favourable cases}}{\text{total number of cases}}$$

2. Types of Events

(i) **Equally likely Events** : The given events are said to be equally likely, if none of them is expected to occur in preference to the other.

(ii) **Independent Events** : Two events are said to be independent if the occurrence of one does not depend upon the other. If a set of events $E_1, E_2, \dots, E_n$ are independent events, then

$$P(E_1 \cap E_2 \cap E_3 \cap \dots \cap E_n) = P(E_1).P(E_2).\dots.P(E_n)$$

(iii) **Mutually Exclusive Events** : A set of events is said to be *mutually exclusive* if occurrence of one of them precludes the occurrence of any of the remaining events.

If a set of events $E_1, E_2, \dots, E_n$ are mutually exclusive events, then

$$P(E_1 \cap E_2 \cap E_3 \cap \dots \cap E_n) = \phi$$

$$\text{then } P(E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots P(E_n)$$

(iv) **Exhaustive Events** : A set of events is said to be *Exhaustive* if the performance of the experiment results in the occurrence of at least one of them.

If a set of Events $E_1, E_2, \dots, E_n$ are exhaustive events, then

$$P(E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n) = 1$$

(v) **Compound Events** : If $E_1, E_2, \dots, E_n$ are mutually exclusive and exhaustive events, then if E is any event,

$$P(E) = \sum_{i=1}^n P(E \cap E_i) = \sum_{i=1}^n P(E_i) \cdot P\left(\frac{E}{E_i}\right), \text{ if } P(E_i) > 0.$$

3. Conditional Probability

The probability of occurrence of an events E_1 , given that E_2 has already occurred is called the conditional probability of occurrence of E_1 on the condition that E_2 has already occurred. It is denoted by $P\left(\frac{E_1}{E_2}\right)$

$$\text{i.e., } P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)}, E_2 \neq \phi$$

4. Bayes's Theorem or Inverse Probability

If $E_1, E_2, \dots, E_n$ are n mutually exclusive and exhaustive events such that

$$P(E_i) > 0, (0 \leq i \leq n)$$

and E is any event, then for $1 \leq k \leq n$,

$$P\left(\frac{E_k}{E}\right) = \frac{P(E_k)P\left(\frac{E}{E_k}\right)}{\sum_{k=1}^n P(E_k)P\left(\frac{E}{E_k}\right)}.$$

5. Important Results

(i) If E_1 and E_2 are arbitrary events, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

(ii) If E_1, E_2, E_3 are three events then

$$\begin{aligned} P(E_1 \cup E_2 \cup E_3) &= P(E_1) + P(E_2) + P(E_3) \\ &\quad - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3) \end{aligned}$$

UNITS – 17 : TRIGONOMETRY

Trigonometric Functions and Identities

1. Measurement of an Angle

- (i) $1^\circ = 60 \text{ minutes} = 60'$ (ii) $1' = 60 \text{ seconds} = 60''$
(iii) Each interior angle of a regular polygon of n sides is equal to
$$\frac{(n-2)\pi}{n}, n > 2.$$

2. Some Important Formulae and Identities

$$\sin(A+B) = \sin A \cos B + \cos A \sin B \quad \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B \quad \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}; \quad \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\tan(45^\circ + A) = \frac{1 + \tan A}{1 - \tan A}; \quad \tan(45^\circ - A) = \frac{1 - \tan A}{1 + \tan A}$$

$$\sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$\cos(A+B) \cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B) \quad 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B) \quad 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2} \quad \sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2} \quad \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2}$$

$$\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A} \quad \cos 2A = \begin{cases} 1 - 2 \sin^2 A \\ \cos^2 A - \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A} \\ 2 \cos^2 A - 1 \end{cases}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} \quad \sin^2 A = \frac{1 - \cos 2A}{2} \text{ and } \cos^2 A = \frac{1 + \cos 2A}{2}$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A;$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

3. Conditional Identities

If $A + B + C = 180^\circ$, then

$$(i) \quad \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$(ii) \quad \cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$$

$$(iii) \quad \sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$(iv) \quad \cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$(v) \quad \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$(vi) \quad \tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$$

4. Trigonometrical Ratios

$$(i) \quad \sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$$

$$(ii) \quad \sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$$

$$(iii) \quad \sin^2 \theta + \cos^4 \theta = \cos^2 \theta + \sin^4 \theta = 1 - \sin^2 \theta \cos^2 \theta$$

$$(iv) \quad \sin^2 \theta + \operatorname{cosec}^2 \theta \geq 2, \cos^2 \theta + \sec^2 \theta \geq 2 \quad \text{and} \quad \sec^2 \theta + \operatorname{cosec}^2 \theta \geq 4$$

5. Some standard Results

$$(i) \quad \tan \theta + \cot \theta = 2 \operatorname{cosec} 2\theta$$

$$(ii) \quad \cot \theta - \tan \theta = 2 \cot 2\theta$$

$$(iii) \quad \sin \theta + \cos \theta \text{ has the same sign as that of } \sin \left(\theta + \frac{\pi}{4} \right)$$

$$(iv) \quad \sin \theta - \cos \theta \text{ has the same sign as that of } \sin \left(\theta - \frac{\pi}{4} \right)$$

(v) Maximum value of $a \sin x + b \cos x$ is $\sqrt{a^2 + b^2}$ and its minimum value is $-\sqrt{a^2 + b^2}$.

(vi) The equation $a \sin x + b \cos x = c$ has real solutions only if $|c| \leq \sqrt{a^2 + b^2}$, i.e., if $c^2 \leq a^2 + b^2$

(vii) $\tan (x_1 + x_2 + x_3 + \dots + x_n) = \frac{S_1 - S_3 + S_5 - \dots}{1 - S_2 + S_4 - \dots}$,

where S_r stands for the sum of the products of $\tan x_1, \tan x_2, \tan x_3, \dots, \tan x_n$ taken r at a time. For example,

$$\tan (A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

$$\tan (A + B + C + D) = \frac{\sum \tan A - \sum \tan A \tan B \tan C}{1 - \sum \tan A \tan B + \tan A \tan B \tan C \tan D}$$

$$\text{In particular, } \tan 4A = \frac{4 \tan A - 4 \tan^3 A}{1 - 6 \tan^2 A + \tan^4 A}$$

(viii) **“m, n” theorem**

If D is a point on the side BC of a triangle ABC

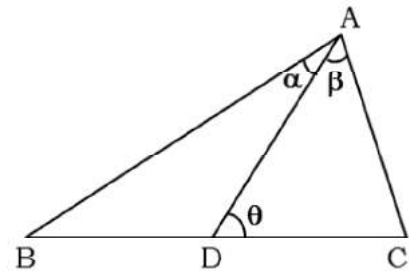
such that $BD : DC :: m : n$,

$\angle ADC = \theta, \angle BAD = \alpha, \angle CAD = \beta$

then

$$(m + n) \cot \theta = m \cot \alpha - n \cot \beta \text{ and}$$

$$(m + n) \cot \theta = n \cot B - m \cot C$$



(ix) $\cos \theta \cos 2\theta \cos 2^2\theta \dots \cos 2^n\theta$

$$= \frac{\sin 2^{n+1}\theta}{2^{n+1} \sin \theta} \text{ for all } n \in \mathbb{N}$$

$$(x) \quad (2 \cos \theta - 1)(2 \cos 2\theta - 1) \dots (2 \cos 2^n \theta - 1) \\ = \frac{2 \cos 2^{n+1} \theta + 1}{2 \cos \theta + 1} \text{ for all } n \in \mathbb{N}$$

$$(xi) \quad \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots \sqrt{2 + 2 \cos 2^n \theta}}}} = 2 \cos \theta$$

for all $n \in \mathbb{N}$, where there are n square root signs on the left hand side.

$$(xii) \quad \sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + \overline{n-1}\beta) \\ = \frac{\sin\left(\frac{\alpha + \alpha + \overline{n-1}\beta}{2}\right) \sin \frac{n\beta}{2}}{\sin\left(\frac{\beta}{2}\right)}, n \in \mathbb{N}$$

$$(xiii) \quad \cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \dots + \cos(\alpha + \overline{n-1}\beta) \\ = \frac{\cos\left(\frac{\alpha + \alpha + \overline{n-1}\beta}{2}\right) \sin \frac{n\beta}{2}}{\sin\left(\frac{\beta}{2}\right)}, n \in \mathbb{N}$$

Trigonometric Equations

$$\sin x = 0 \Leftrightarrow x = n\pi, n \in \mathbb{I}$$

$$\cos x = 0 \Leftrightarrow x = (2n+1)\frac{\pi}{2}, n \in \mathbb{I}$$

$$\tan x = 0 \Leftrightarrow x = n\pi, n \in \mathbb{I}$$

$$\sin x = \sin \alpha \Leftrightarrow x = n\pi + (-1)^n \alpha, n \in \mathbb{I}$$

$$\cos x = \cos \alpha \Leftrightarrow x = 2n\pi \pm \alpha, n \in \mathbb{I}$$

$$\tan x = \tan \alpha \Leftrightarrow x = n\pi \pm \alpha, n \in \mathbb{I}$$

$$\sin^2 x = \sin^2 \alpha \Leftrightarrow x = n\pi \pm \alpha, n \in \mathbb{I}$$

$$\cos x = \cos \alpha \Leftrightarrow x = \pi \pm \alpha, \quad x \in \mathbb{I}$$

$$\tan^2 x = \tan^2 \alpha \Leftrightarrow x = n\pi \pm \alpha, \quad n \in \mathbb{I}$$

Properties and Solution of Triangles

1. Relations between the Sides and Angles of a Triangle

(i) Sine Formulae

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, where R is the circumradius of the triangle.

or $a = k \sin A, b = k \sin B, c = k \sin C$ where k is some non-zero constant.

or $\sin A = \frac{a}{k}, \sin B = \frac{b}{k}, \sin C = \frac{c}{k}$, k being some non-zero constant.

(ii) Cosine Formulae

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{c^2 + a^2 - b^2}{2ca} \text{ and } \sin C = \frac{a^2 + b^2 - c^2}{2ab}$$

(iii) Projection Formulae

$$a = b \cos C + c \cos B; b = c \cos A + a \cos C; c = a \cos B + b \cos A$$

(iv) Napier's Analogy

$$\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}; \tan \frac{C-A}{2} = \frac{c-a}{c+a} \cot \frac{B}{2}; \tan \frac{A-B}{2} = \frac{a-b}{a+b} \cot \frac{C}{2}$$

(v) Semi-sum Formulae

$$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}; \sin \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}; \sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

$$\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}; \quad \cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}; \quad \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}; \quad \tan \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}; \quad \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

where, $s = (a + b + c) / 2$

$$\Delta = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B, \text{ where } \Delta = \text{Area of } \Delta ABC$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} \quad (\text{Hero's formula})$$

2. Circumradius (R) and Inradius (r) Formulae

$$(i) \quad R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C} = \frac{abc}{4 \Delta}$$

$$(ii) \quad r = \frac{\Delta}{s} = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2}$$

$$(iii) \quad r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

Inverse Trigonometric Functions

Important Formulae

$$y = \sin^{-1}x \text{ iff } x = \sin y, |x| \leq 1, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$y = \cos^{-1}x \text{ iff } x = \cos y, |x| \leq 1, y \in [0, \pi]$$

$$y = \tan^{-1}x \text{ iff } x = \tan y, x \in \mathbb{R}, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$y = \cot^{-1}x \text{ iff } x = \cot y, x \in \mathbb{R}, y \in (0, \pi)$$

$$y = \sec^{-1}x \text{ iff } x = \sec y, |x| \geq 1, y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

$$y = \operatorname{cosec}^{-1}x \text{ iff } x = \operatorname{cosec} y, |x| \geq 1, y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$$

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, |x| \leq 1 \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, x \in \mathbb{R}$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, |x| \geq 1 \quad \sin^{-1}(-x) = -\sin^{-1} x, |x| \leq 1$$

$$\cos^{-1}(-x) = \pi - \cos^{-1} x, |x| \leq 1 \quad \tan^{-1}(-x) = -\tan^{-1} x, x \in \mathbb{R}$$

$$\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x, |x| \geq 1 \quad \sec^{-1}\left(\frac{1}{x}\right) = \cos^{-1} x, 0 < |x| \leq 1$$

$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x, |x| \geq 1 \quad \operatorname{cosec}^{-1}\left(\frac{1}{x}\right) = \sin^{-1} x, 0 < |x| \leq 1$$

$$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x, x > 0 \quad \sin(\cos^{-1} x) = \cos(\sin^{-1} x) = \sqrt{1-x^2}, |x| \leq 1$$

$$\sec(\operatorname{cosec}^{-1} x) = \operatorname{cosec}(\sec^{-1} x) = \frac{|x|}{\sqrt{x^2-1}}; |x| > 1$$

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1 \quad \tan^{-1}\left(\frac{1+x}{1-x}\right) = \frac{\pi}{4} + \tan^{-1} x, x < 1$$

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{\pi}{4} - \tan^{-1} x, x > -1 \quad \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2 \tan^{-1} x, x \in \mathbb{R}$$

$$\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2 \tan^{-1} x, x \geq 0 \quad \tan^{-1}\left(\frac{2x}{1-x^2}\right) = 2 \tan^{-1} x, |x| < 1$$

$$2 \sin^{-1} x = \sin^{-1}\left(2 \sqrt{1-x^2}\right), |x| \leq \frac{1}{\sqrt{2}} \quad 2 \cos^{-1} x = \cos^{-1}(2x^2-1), x \in [0, 1]$$

$$3 \sin^{-1} x = \sin^{-1}(3x-4x^3), |x| \leq \frac{1}{2} \quad 3 \cos^{-1} x = \cos^{-1}(4x^3-3x), \frac{1}{2} \leq x \leq 1$$

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

for those values of x and y in $[-1, 1]$ for which LHS lies in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

where x, y are real numbers in $[-1, 1]$ such that LHS lies in the interval $[0, \pi]$.

- (i) $\sin^{-1} x + \sin^{-1} y = \pi$ iff $x = y = 1$
- (ii) $\sin^{-1} x + \sin^{-1} y = -\pi$ iff $x = y = -1$
- (iii) $\cos^{-1} x + \cos^{-1} y = 0$ iff $x = y = 1$
- (iv) $\cos^{-1} x + \cos^{-1} y = 2\pi$ iff $x = y = -1$
- (v) $\sin^{-1} \left(\frac{2x}{1+x^2} \right) = 2 \cot^{-1} x = 2 \tan^{-1} \left(\frac{1}{x} \right)$ for $x \geq 1$
- (vi) $\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right)$, where $xy > 1, x > 0, y > 0$.

UNITS – 18 : MATHEMATICAL REASONING

- (i) $p \wedge q$ is true if p and q are both true.
- (ii) $p \vee q$ is false if p and q are both false.
- (iii) If p is true, then $\sim p$ is false.
If p is false, then $\sim p$ is true.
- (iv) $p \Rightarrow q$ is true in all cases except when p is true and q is false.
- (v) $p \Leftrightarrow q$
 - (a) $p \Leftrightarrow q$ is true if both p and q have same truth value.
 - (b) $p \Leftrightarrow q$ is false if p and q have opposite truth value.
- (vi) If p is any statement, t is tautology and c is contradiction, then :
 - (a) $p \vee t \equiv t$ (b) $p \wedge t \equiv p$
 - (c) $p \vee c \equiv p$ (d) $p \wedge c \equiv c$

Complement Laws :

- (a) $p \vee (\sim p) \equiv t$ (b) $p \wedge (\sim p) \equiv c$
(c) $\sim t \equiv c$ (d) $\sim c \equiv t$

Involution Laws :

- (a) $\sim(\sim p) = p$

DeMorgan's Laws :

If p and q are any two statements, then :

- (a) $\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$ (b) $\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$

Laws of Contrapositive :

$$p \Rightarrow q \equiv \sim q \Rightarrow \sim p$$

- (vii) If a compound statement contains variables t (tautology) and c (contradiction), then its dual is obtained by replacing t by c and c by t in addition to the replacement of $\vee$ and $\wedge$ by $\wedge$ and $\vee$ respectively.

Types of Statements :

- i) If a statement is always true, then the statement is called "tautology."
- ii) If a statement is always false, then the statement is called "contradiction."
- iii) If a statement is neither tautology nor a contradiction, then it is called "contingency."

Converse, Contrapositive, Inverse of a Statement

If $p \rightarrow q$ is a hypothesis, then

- i) Converse $q \rightarrow p$.
- ii) Contrapositive $\sim q \rightarrow \sim p$.
- iii) Inverse $\sim p \rightarrow \sim q$.